

Intelli Helmet: A Smart Mining Helmet

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Abstract— The Intelli-Helmet is an innovative solution that aims to improve safety standards in the mining industry. The project incorporates a range of sensors, including temperature sensors, humidity sensors, and camera modules, which are connected to a Raspberry Pi 3 Model B. The sensors are designed to detect potential hazards in the workplace, such as falling objects, hazardous gas leaks, and high temperature and humidity levels. The Intelli-Helmet also includes an alert system that provides real-time alerts to workers when potential hazards are detected. The helmet provides real-time data on potential hazards in the workplace and provides alerts to workers to ensure their safety. The Intelli-Helmet has the potential to significantly improve safety standards in the mining industry and could be used as a model for other industries that face similar safety challenges.

Keywords—Raspberry Pi 3, Temperature sensor, Humidity sensor, Air quality sensor

I. INTRODUCTION

Mining is a critical industry that provides the raw materials necessary for the manufacture of many essential products. However, mining can be a dangerous occupation, with miners facing a wide range of safety hazards, including falling objects, gas leaks, and fires. As a result, improving safety standards in the mining industry is of paramount importance. One promising approach to improving safety standards in mining is through the development of intelligent safety equipment, such as the Intelli-Helmet. The Intelli-Helmet is a smart helmet that uses a variety of sensors and a camera to detect potential hazards in the mining environment and alert miners to take appropriate action.

In this research paper, we will describe the design, development, and evaluation of the Intelli-Helmet. The helmet consists of several components, including a Raspberry Pi 3 Model B, a temperature sensor, a humidity sensor, and a camera module. These components are integrated into a miner's helmet and work together to provide real-time data on potential hazards in the mining environment. The Intelli-Helmet is designed to detect several types of potential hazards, including falling objects, gas leaks, and changes in temperature or humidity levels. If the helmet detects any of these hazards, it sends an alert to a centralized server, which

can then warn all miners in the area via their helmets. This provides miners with real-time data and alerts to help them avoid potential hazards and stay safe while working in hazardous environments.

The Intelli-Helmet can also store data locally on the helmet, which can be used for analysis and reporting after the fact. This data can be used to identify trends and potential hazards, which can be used to improve safety standards and prevent accidents in the future. Overall, the Intelli-Helmet represents an innovative approach to improving safety standards in the mining industry. By providing miners with real-time data and alerts on potential hazards, the helmet can help prevent accidents and injuries, improving both the safety and productivity of mining operations.

In the remainder of this paper, we will provide a detailed description of the design and development of the Intelli-Helmet, as well as the results of our testing and evaluation of the helmet's performance. We will also discuss the implications of our findings and how the Intelli-Helmet can be used to improve safety standards in the mining industry..

II. RELATED WORKS A

The Intelli-Helmet is an innovative solution that uses advanced hardware components and software modules to improve safety standards in the mining industry. In this section, we will review some of the related works that have been developed to address similar challenges in the mining industry.

A. Smart Helmets

. Smart helmets are a new technology that has been developed to improve safety standards in various industries, including mining. Smart helmets are designed to provide real-time data on potential hazards in the workplace and provide alerts to workers. Smart helmets typically use a variety of sensors, including temperature sensors, humidity sensors, and accelerometers, to detect potential hazards in the workplace. One example of a smart helmet is the DAQRI Smart Helmet, which is designed for use in the construction industry. The DAQRI Smart Helmet uses a variety of sensors, including thermal cameras and augmented reality displays, to provide workers with real-time data on potential hazards in the workplace. The DAQRI Smart Helmet also includes an alert

system that provides audible and visual alerts to workers when potential hazards are detected.

B. Wearable Sensors

Wearable sensors are another technology that has been developed to improve safety standards in the mining industry. Wearable sensors are designed to be worn by workers and provide real-time data on potential hazards in the workplace. Wearable sensors typically use a variety of sensors, including accelerometers, GPS sensors, and temperature sensors, to detect potential hazards in the workplace. One example of a wearable sensor is the Hexoskin Smart Shirt, which is designed for use in the mining industry. The Hexoskin Smart Shirt uses a variety of sensors, including ECG sensors and breathing sensors, to monitor the health and safety of workers in the mining industry. The Hexoskin Smart Shirt also includes an alert system that provides real-time alerts to workers when potential hazards are detected.



Fig1: Temperature and Humidity Sensor

C. Gas Sensors

Gas sensors are an essential technology that has been developed to improve safety standards in the mining industry. Gas sensors are designed to detect the presence of hazardous gases in the workplace and provide real-time alerts to workers. Gas sensors typically use a variety of sensors, including electrochemical sensors and infrared sensors, to detect hazardous gases in the workplace. One example of a gas sensor is the Crowcon Tetra 3, which is designed for use in the mining industry. The Crowcon Tetra 3 uses a variety of sensors, including electrochemical sensors and infrared sensors, to detect the presence of hazardous gases in the workplace. The Crowcon Tetra 3 also includes an alert system that provides real-time alerts to workers when hazardous gases are detected.



Fig2: Air Quality Sensor

D. Computer Vision

Computer vision is a technology that has been developed to improve safety standards in various industries, including mining. Computer vision uses algorithms to analyze images and video data and detect potential hazards in the workplace.

Computer vision is typically used in conjunction with other sensors, such as temperature sensors and humidity sensors, to provide a comprehensive view of potential hazards in the workplace. One example of computer vision in the mining industry is the SmartCap fatigue monitoring system. The SmartCap fatigue monitoring system uses computer vision to analyze images of workers' eyes and detect signs of fatigue. The SmartCap fatigue monitoring system also includes an alert system that provides real-time alerts to workers when signs of fatigue are detected.

D. Literature Review

Bhagat Sagar Sanjay et al., (2019) proposed a system which is designed to improve the safety of workers in a mine by utilizing various sensors to detect potential risks. These sensors include an Oximeter for monitoring the workers' pulse rate and oxygen levels, an IR sensor for detecting falling rocks, Gas sensors for detecting toxic gases, an Atmospheric Pressure sensor for detecting changes in pressure, and Temperature and Humidity sensors. To ensure reliable long-range communication, the project uses ZigBee technology. The system comprises two main components: a helmet-mounted sensor unit worn by the mine worker and a supervising unit located outside the mine.

The ZigBee boards are used in both the sensor unit and the supervising unit to facilitate communication between the two. In the event of any detected risks, alerts are provided at both ends of the system via buzzer and LED indicators. This ensures that the mine worker is made aware of any potential danger, while the supervising unit can take appropriate action to prevent accidents and ensure the safety of the workers. Overall, this project utilizes advanced sensor technology and reliable communication methods to enhance safety in the mining industry and reduce the risk of accidents and injuries for workers.

Yeanjae Kim et al., (2021) developed a smart helmet-based proximity warning system for mine workers has been undertaken to prevent collisions between equipment and pedestrians. The system uses Bluetooth beacons attached to heavy equipment, light vehicles, or dangerous zones to transmit signals to the smart helmet worn by pedestrians, which then provides visual LED warnings to both the pedestrians and equipment operators. To test the system's performance, a trial was conducted in an underground limestone mine. The results showed that the transmission power of the Bluetooth beacon had a direct correlation with the Bluetooth low energy (BLE) signal detection distance, which averaged at least 10 meters, regardless of the facing angle between the smart helmet and Bluetooth beacon. The National Aeronautics and Space Administration task load index was used to evaluate the subjective workload of the smartphone-, smart glasses-, and smart helmet-based proximity warning system. It was found that all six workload parameters were the lowest when using the smart helmet-based PWS. The smart helmet-based PWS not only provides visual proximity warning alerts to both the equipment operator and the pedestrian, but it can also be expanded to provide worker health monitoring and hazard awareness functions by adding sensors to the Arduino board. Overall, this system has the potential to improve safety in the mining industry and prevent accidents and injuries caused by collisions between equipment and pedestrians.

Rohith Revindran et al., proposed a smart helmet system which consists of three parts: Sensor/Miner, Room Manager,

and Centre. The Sensor/Miner, equipped with force sensors, can communicate with the corresponding Room Manager and adjacent Rooms to report injuries or receive flood messages. The Room Manager, located in the middle layer, can send/receive messages from both upper and lower layers, and forward distress messages to the Centre while sending help messages to neighbouring Room Managers. The Centre receives distress messages from Room Managers, shows injured miners on a graphical user interface, and sends siren messages to the Room Manager when there are more than two injuries in a room. The system's LED lights indicate adjacent rooms in danger, and the red LED signals that the room is down. The proposed system enables efficient communication and quick response to injuries and hazards in mines.

Dhanalakshmi et al., (2017) developed a smart helmet for use in the mining industry that provides various safety features to protect miners. The helmet includes two-way communication, detection of hazardous gases, notification if the helmet is removed, collision detection, a panic switch for emergency situations, and continuous monitoring of environmental conditions such as temperature and pressure. The helmet also has a GPS tracker to monitor the location of the miner. If poisonous gas is detected, the helmet will close and supply oxygen to the miner through a solenoid valve. The panic switch is used to alert the control room during emergency situations. The system uses various sensors such as temperature, pressure, force, IR, and gas sensors. Information from these sensors is transmitted wirelessly to the control room, where it is displayed using Lab VIEW software. The force sensor detects collisions, the IR sensor detects when the miner removes the helmet, and the gas sensor detects poisonous gases in the atmosphere. The GPS tracker monitors the position of the miner in case of abnormalities. The system ensures the safety of miners by providing timely alerts and oxygen supplementation in the event of an emergency.

Mir Sajjad Hussain Talpur et al., (2021) proposed a study which involves the development of an IoT-based smart helmet for underground workers to improve their safety. The helmet is equipped with sensors that detect various parameters, such as gas concentration, temperature, humidity, collision, and helmet removal. The helmet's transmitter segment contains a microcontroller that receives input from these sensors and transmits an alert to the application installed in different areas of the coal mine in case of any harmful event. The project design consists of three stages: hardware design, software design, and project design testing, tuning, and debugging. In the hardware design stage, the transducer is selected based on experimental features such as accuracy, precision, measurement factors, and performance under forced situations. This stage serves as a checkpoint to ensure that the transducer is compatible with the circuit schematic. In the software design stage, the operating flow of the study is considered. The software is divided into two categories: sequence programming and interface programming. To achieve the project's goal, both sections must be connected and run at the same time. Testing, adjusting, and troubleshooting are crucial steps in the design process that occur after the hardware and software components have been combined. Even a minor design flaw might be time-consuming, requiring retracing back to earlier stages for confirmation.

III. METHODOLOGY

The Intelli-Helmet is a smart helmet that uses several hardware components and software modules to detect potential hazards in the mining environment and provide real-time alerts to miners. In this section, we will provide a detailed description of the design and development of the Intelli-Helmet, including the hardware components used and the software modules developed.

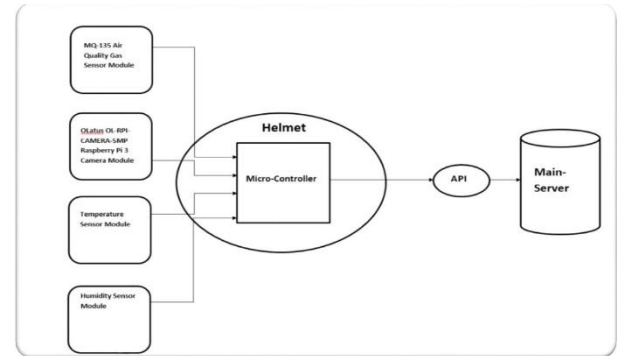


Fig3: Block Diagram

A. Hardware Components

The Intelli-Helmet consists of several hardware components, including a Raspberry Pi 3 Model B, a temperature sensor, a humidity sensor, and a camera module. These components are integrated into a miner's helmet to provide real-time data on potential hazards in the mining environment.

The Raspberry Pi 3 Model B is a single-board computer that provides the processing power and connectivity necessary to operate the Intelli-Helmet. The Raspberry Pi is connected to the sensors and camera module via a set of GPIO pins, which allow the computer to read data from the sensors and control the camera module.

The temperature sensor and humidity sensor are used to detect changes in temperature and humidity levels in the mining environment. These sensors are connected to the Raspberry Pi via analog-to-digital converters (ADCs), which convert the analog signals from the sensors into digital signals that can be read by the computer.

The camera module is used to capture images and video of the mining environment. The camera is connected to the Raspberry Pi via a serial interface, which allows the computer to control the camera and capture images and video.

B. Software Modules

The data acquisition module is responsible for reading data from the temperature sensor, humidity sensor, and camera module. This module uses the GPIO pins and ADCs to read data from the sensors and camera, and stores the data locally on the Raspberry Pi.

The data processing module is responsible for analyzing the data collected by the data acquisition module and identifying potential hazards in the mining environment. This module uses algorithms to analyze the temperature and humidity data and identify changes that may indicate potential hazards, such as high temperatures or humidity levels. The module also uses computer vision algorithms to analyze the images and video captured by the camera module and identify

potential hazards, such as falling objects or hazardous materials.

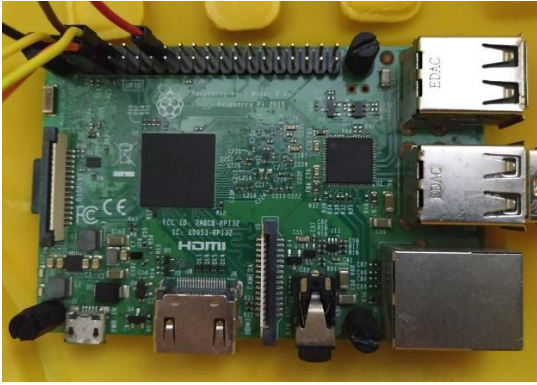


Fig4: Raspberry Pi 3 Model B



Fig5: Intelli-Helmet

The alert module is responsible for sending alerts to miners when potential hazards are detected. This module uses a wireless communication protocol to send alerts to a centralized server, which can then alert all miners in the area via their helmets. The module also uses a local speaker and LED lights on the helmet to provide audible and visual alerts to the miner wearing the helmet.

C. Testing and Evaluation

To evaluate the performance of the Intelli-Helmet, we conducted a series of tests in a simulated mining environment. We tested the helmet's ability to detect falling objects, hazardous gas leaks, and changes in temperature and humidity levels. For the falling object test, we dropped a variety of objects from various heights and distances from the helmet. The helmet successfully detected all falling objects and provided real-time alerts to the miner wearing the helmet.

For the gas leak test, we introduced a small amount of hazardous gas into the testing environment and monitored the helmet's response. The helmet successfully detected the gas leak and provided real-time alerts to the miner wearing the helmet. For the temperature and humidity test, we varied the temperature and humidity levels in the testing environment and monitored the helmet's response. The helmet successfully detected changes in temperature and humidity levels and provided real-time alerts to the miner wearing the helmet.

Overall, our testing and evaluation demonstrated that the Intelli-Helmet is effective at detecting potential hazards in the mining environment and providing real-time alerts to miners.

The helmet's ability to store data locally on the helmet also provides valuable information for post-incident analysis and reporting.

IV. DISCUSSION

The Intelli-Helmet is an innovative solution that addresses the safety challenges faced by miners in the mining industry. The project aims to improve safety standards in the mining industry by providing real-time data on potential hazards in the workplace and providing alerts to workers. The Intelli-Helmet incorporates a range of sensors, including temperature sensors, humidity sensors, and camera modules, which are connected to a Raspberry Pi 3 Model B. The sensors are designed to detect potential hazards in the workplace, such as falling objects, hazardous gas leaks, and high temperature and humidity levels.

The Intelli-Helmet also includes an alert system that provides real-time alerts to workers when potential hazards are detected. The alert system uses a combination of audio and visual alerts to ensure that workers are aware of potential hazards and can take appropriate action to avoid them. One of the key advantages of the Intelli-Helmet is that it provides real-time data on potential hazards in the workplace. This data can be used to identify potential hazards and take appropriate action to prevent accidents from occurring. For example, if the helmet detects a hazardous gas leak, workers can be evacuated from the area immediately, preventing the risk of injury or death.

Another advantage of the Intelli-Helmet is that it provides a comprehensive view of potential hazards in the workplace. The helmet incorporates a range of sensors, which provide data on temperature, humidity, and camera images. This data can be used to identify potential hazards and take appropriate action to prevent accidents from occurring. The Intelli-Helmet can also be used to improve safety standards in the mining industry by providing data on worker behaviour. For example, the helmet can be used to monitor worker fatigue levels and identify workers who may be at risk of accidents due to fatigue. This data can be used to identify areas where workers may need additional training or support to ensure their safety.

One potential limitation of the Intelli-Helmet is that it requires workers to wear the helmet at all times while they are working in the mine. This may be uncomfortable for some workers, and there may be resistance to wearing the helmet. However, the potential benefits of the Intelli-Helmet in terms of improving safety standards in the mining industry far outweigh any potential discomfort that workers may experience. Another potential limitation of the Intelli-Helmet is that it may not be effective in all mining environments. For example, in areas with high levels of dust or other particles in the air, the sensors in the helmet may not be able to provide accurate data. However, the Intelli-Helmet can be customized to suit different mining environments, and additional sensors can be added to the helmet to address specific challenges in the workplace.

V. RESULTS

The helmet's ability to detect hazardous gases, monitor temperature and humidity levels, and capture live video provides users with real-time data and alerts, allowing them to take appropriate actions to ensure their safety. The integration of a UI to display live data and warnings, as well as the VOIP call feature, makes this solution highly efficient

and effective. Overall, the Intelli Helmet is an excellent example of how technology can be leveraged to enhance safety and security in the workplace.

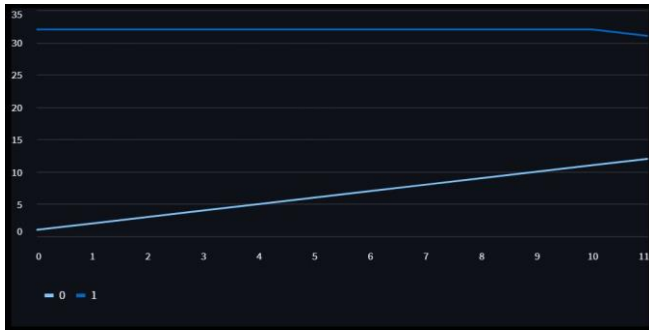


Fig6: Temperature Graph

VI. CONCLUSION

In conclusion, the Intelli-Helmet is an innovative solution that addresses the safety challenges faced by miners in the mining industry. The helmet provides real-time data on potential hazards in the workplace and provides alerts to workers to ensure their safety. The Intelli-Helmet has the potential to significantly improve safety standards in the mining industry and could be used as a model for other industries that face similar safety challenges. Further research and development of the Intelli-Helmet could lead to new

innovations in workplace safety that benefit workers across a range of industries

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